

Recap

$$H = E[I(x)] = \sum_x p(x) I(x) = - \sum_x p(x) \log p(x)$$

- Entropy

$$H(x, y) = - \sum_{x, y} p(x, y) \cdot \log p(x, y)$$

- Joint Entropy, Conditional Entropy

$$H(Y|X) = \sum_x p(x) H(Y|X=x)$$

an informative feature leads to high $I(X_i, Y)$

average conditional entropy classification

- Mutual Information

$$I(X_i, Y) = H(Y) - H(Y|X_i)$$

reduction in uncertainty of Y given we observed feature X_i
 X_i has a lot of information (weight in cat/dog)

- Cross Entropy

low inefficiency

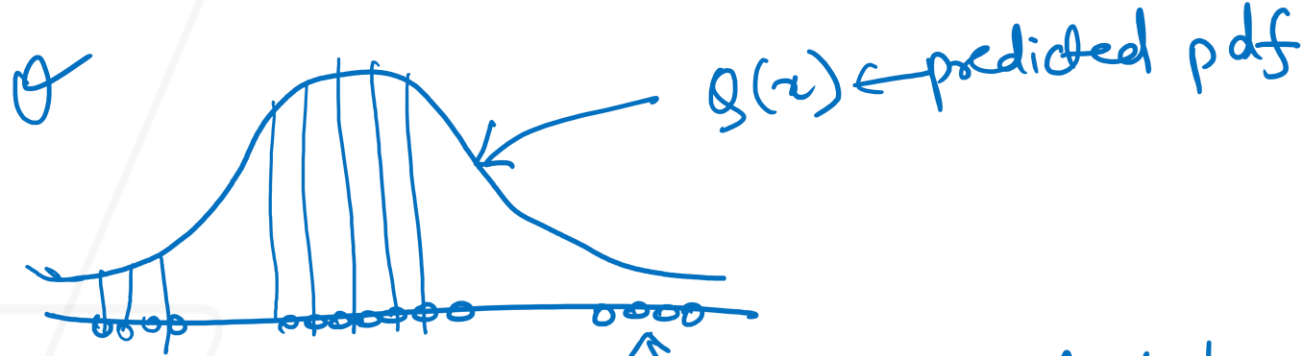
we are encoding samples from $P(x)$ by actually following $Q(x)$

$$H(Y|X_i) \downarrow \quad H(Y) - H(Y|X_i) \uparrow$$

Optimization

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$$E[a+b+c] = E[a] + E[b] + E[c]$$

$$E[g(x)] = \sum_x g(x) \cdot p(x)$$

$$L(\theta|x) = \prod_{i=1}^n f(x^{(i)}|\theta)$$

maximize

$$\log L(\theta|x) = \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

minimize

$$l(\theta|x) = - \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

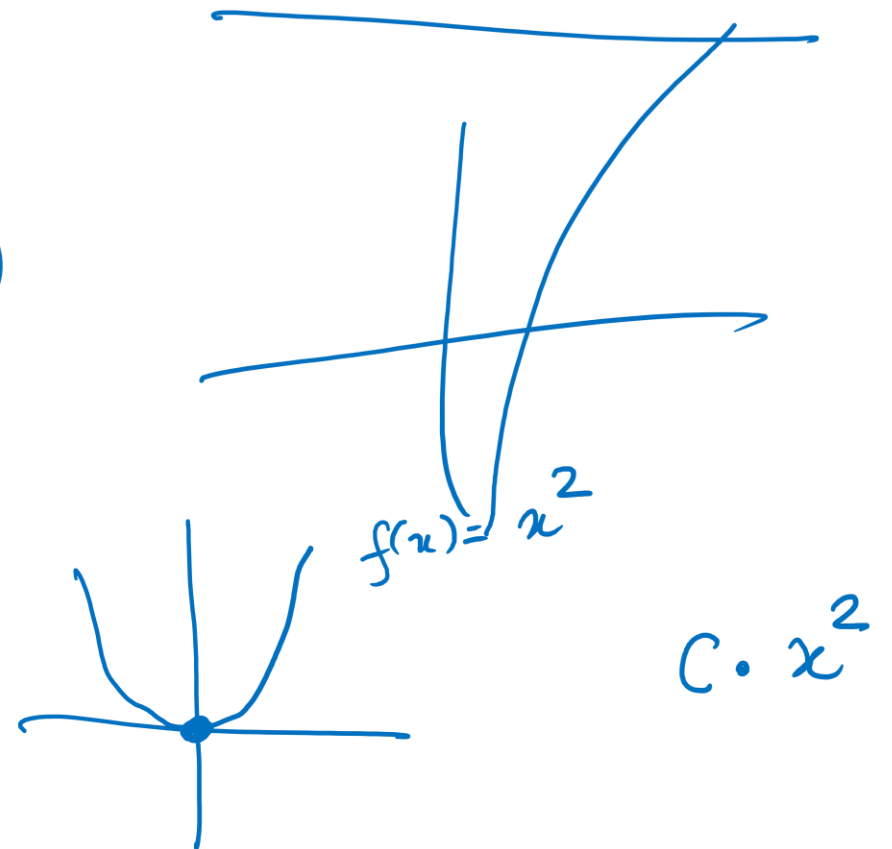
$$= - \frac{1}{n} \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

$$= - E[\log Q(x)]$$

$$= - E[\log P(x) - \log P(x) + \log Q(x)]$$

$$= - E[\log P(x)] - E\left[\log \frac{Q(x)}{P(x)}\right]$$

$$H(P) = - \sum_x p(x) \log P(x) = - \sum_x p(x) \log \frac{Q(x)}{P(x)}$$



①

$$\min l(\theta|x) = \underbrace{-\sum_x p(x) \log p(x)}_{H[p]} - \underbrace{\sum_x p(x) \log \frac{q(x)}{p(x)}}_{KL[p][q]}$$

likelihood minimization = Min. $H[p] + KL[p][q]$

Minimizing CE is the same as minimizing -ve log likelihood.

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(P) + KL[P][Q]$$

Handwritten annotations:

- $H(p, q)$ is circled and labeled "loss function" and "CE".
- The sum $-\sum_{x \in \mathcal{X}} p(x) \log q(x)$ is boxed and labeled "objective function".
- The term $KL[P][Q]$ is bracketed and labeled "non negative".
- The sum $\sum_x p(x) \log \frac{1}{q(x)}$ is written below the main equation, with an arrow pointing to the $\log$ term in the boxed expression.

This is because:

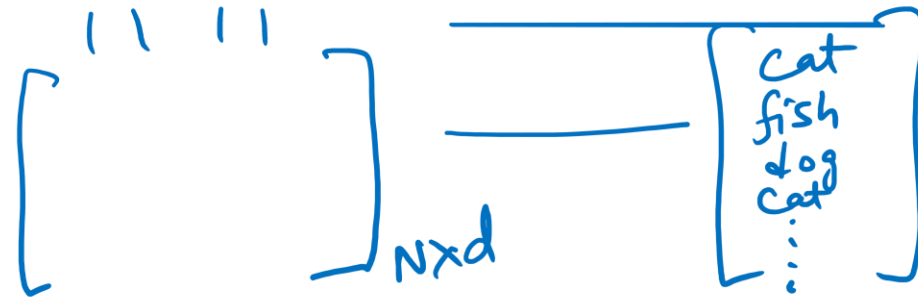
$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[\log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

$$H(p, q) = - \sum_x p(x) \log q(x).$$

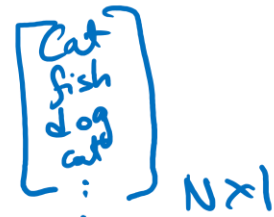
Labeling target values: Label encoding (ordinal) and One-hot encoding

• Input Matrix (X):



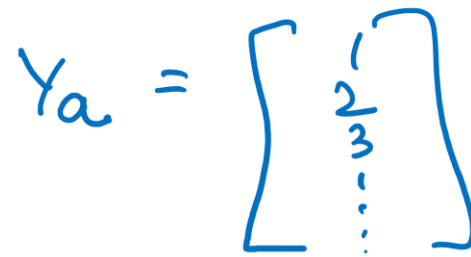
Multiclass classification

• Target Labels (Y):

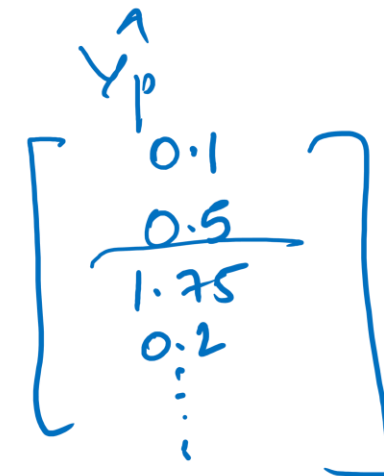


• Label Encoding (Ordinal):

cat $\rightarrow$ 1, fish $\rightarrow$ 2, dog $\rightarrow$ 3



model

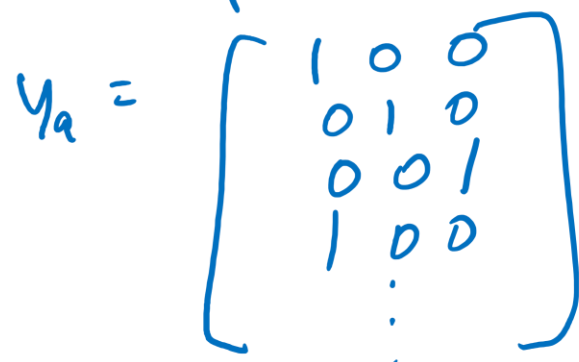


tshirt sizes

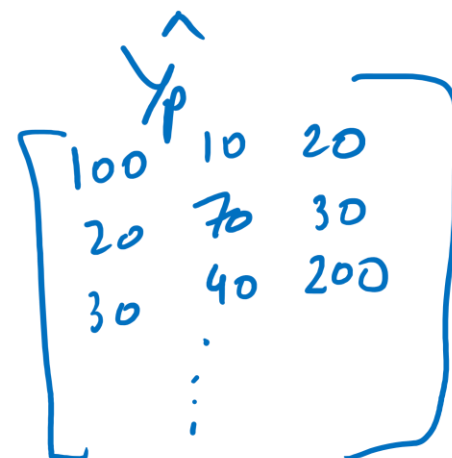
• One-Hot Encoding:

Cat $\rightarrow$ [1 0 0], fish $\rightarrow$ [0 1 0], dog $\rightarrow$ [0 0 1]

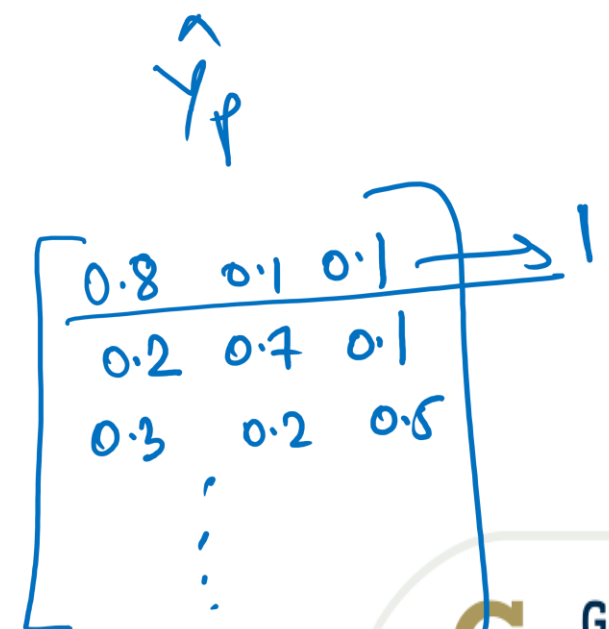
sums up to 1
pm f

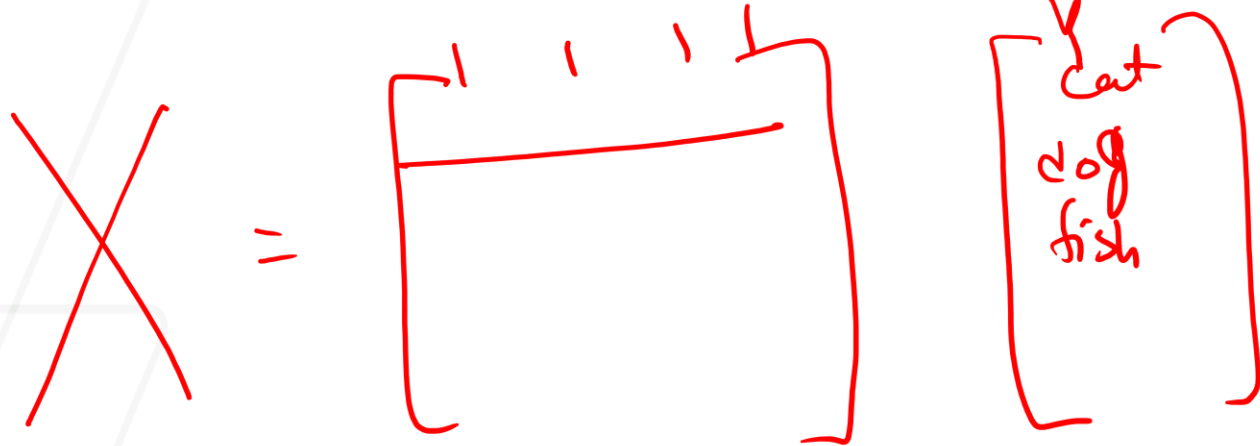


model

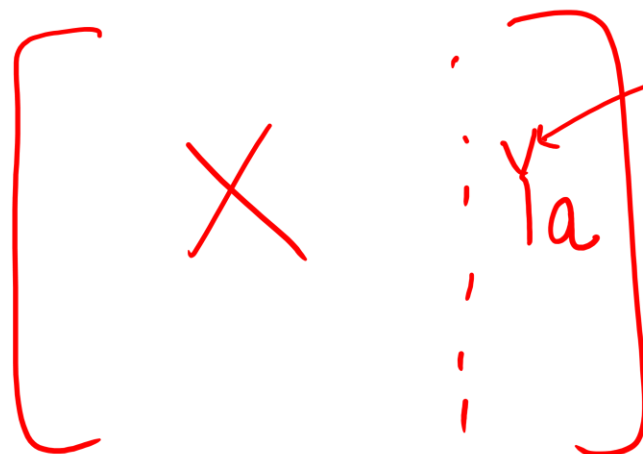


Softmax





Training data



actual labels

Model

use first set of parameters that it assumes



predicted target values

How Y_a and $\hat{Y}_p$ vary

as close as possible

Trained Model $\rightarrow$ a new datapoint $\rightarrow \hat{Y}_p$

$f(x)$
Objective

maximize or minimize



Label encoding (One-hot encoding) Loss Computation

Minimize Objective Function:

$$\min \sum_{i=1}^n \left\| \mathbf{y}_a^{(i)} - \mathbf{y}_p^{(i)T} \right\|_2^2$$

MSE

1st data point

$$\mathbf{y}_a = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$
$$\hat{\mathbf{y}}_p = \begin{bmatrix} 0.8 & 0.1 & 0.1 \end{bmatrix}$$

$$(1 - 0.8)^2 + (0 - 0.1)^2 + (0 - 0.1)^2 = (0.2)^2$$

In a perfect world: $\hat{\mathbf{y}}_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$

$$\rightarrow 0 + 0 + 0 + \dots + n = 0$$

Maximize Objective Function:

$$\max \sum_{i=1}^n \mathbf{y}_a^{(i)} \cdot \hat{\mathbf{y}}_p^{(i)T}$$

$$\min - \sum_{i=1}^n \mathbf{y}_a^{(i)} \cdot \hat{\mathbf{y}}_p^{(i)T}$$

$$0.8 \times 1 + 0 \times 0.1 + 0 \times 0.1 = 0.8$$

In a perfect world = 1

$$1 + 1 + 1 + \dots + n = n$$

Interpretation:

- Minimum squared Error works but isn't ideal for classification.
- Minimize negative sum of dot product is also good but Cross-Entropy / NLL is better.

Why Cross Entropy is better than dot product

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

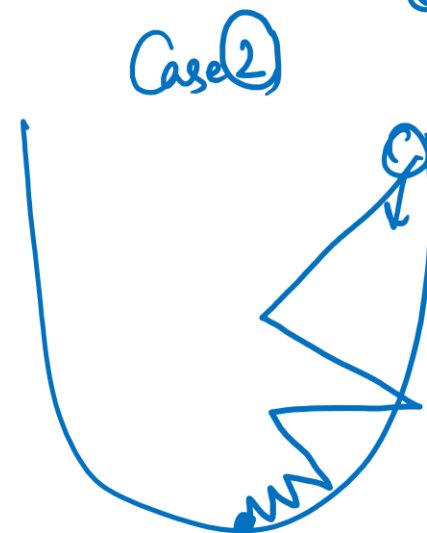
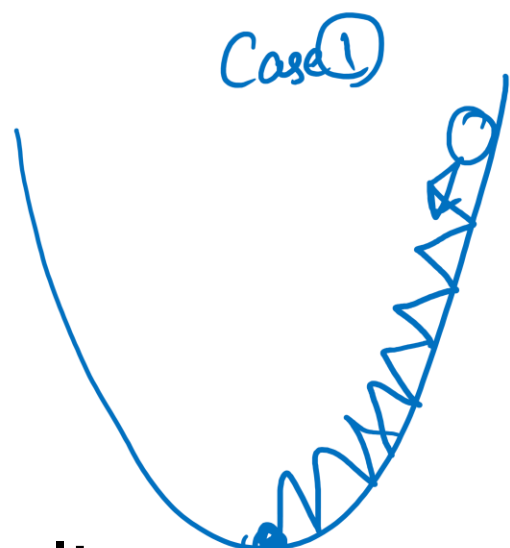
$\uparrow$ $y_a^{\{i\}}$ $\uparrow$ $\hat{y}_p^{\{i\}}$ $\uparrow$ $\text{-ve log likelihood}$ $\uparrow$ $\text{CE} =$

$$= \min_{i=1}^n y_a^{\{i\}} \cdot \log \hat{y}_p^{\{i\}T} \quad \text{Case ②}$$

$$\min_{i=1}^n y_a^{\{i\}} \cdot \hat{y}_p^{\{i\}T} \quad \text{Case ①}$$



Log really really penalizes our objective function



Convergence is quick

Logarithmic penalty encourages big update leading to faster convergence

Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to Q instead of P .

KL Divergence is
a **KIND OF**
distance
measurement

$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is
concave or
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

By Jensen Inequality

So $\mathbf{KL}[P||Q] \geq 0$. Equality iff $P = Q$

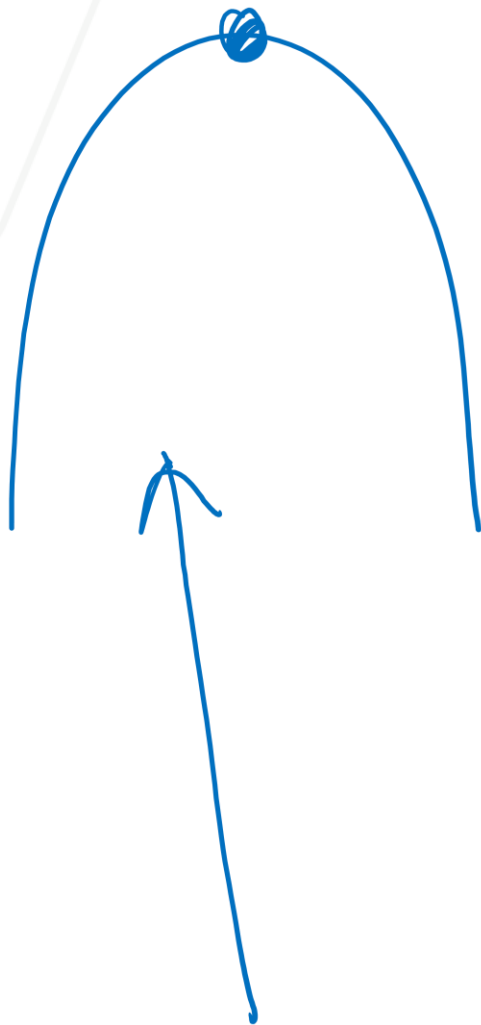
When $P = Q$, $KL[P||Q] = 0$

Concave Function: Jensen Inequality

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$$

$$f(x) = -x^2$$

$$f''(x) < 0 \rightarrow \text{concave}$$



Strictly concave functions have global maximum

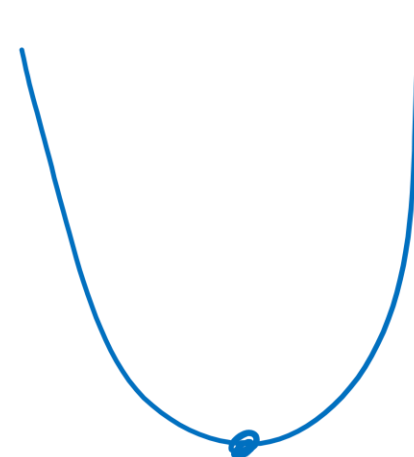
Convex Function: Jensen Inequality

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

$$f(x) = x^2$$

$$x = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$p(x) = \begin{bmatrix} 1/2 & 1/2 \end{bmatrix}$$



$$\begin{aligned} \mathbb{E}[f(x)] &= \sum_x f(x) \cdot p(x) = f(x=1)p(x=1) + f(x=2)p(x=2) \\ &= 1 \cdot 1/2 + 4 \cdot 1/2 = 2.5 \end{aligned}$$

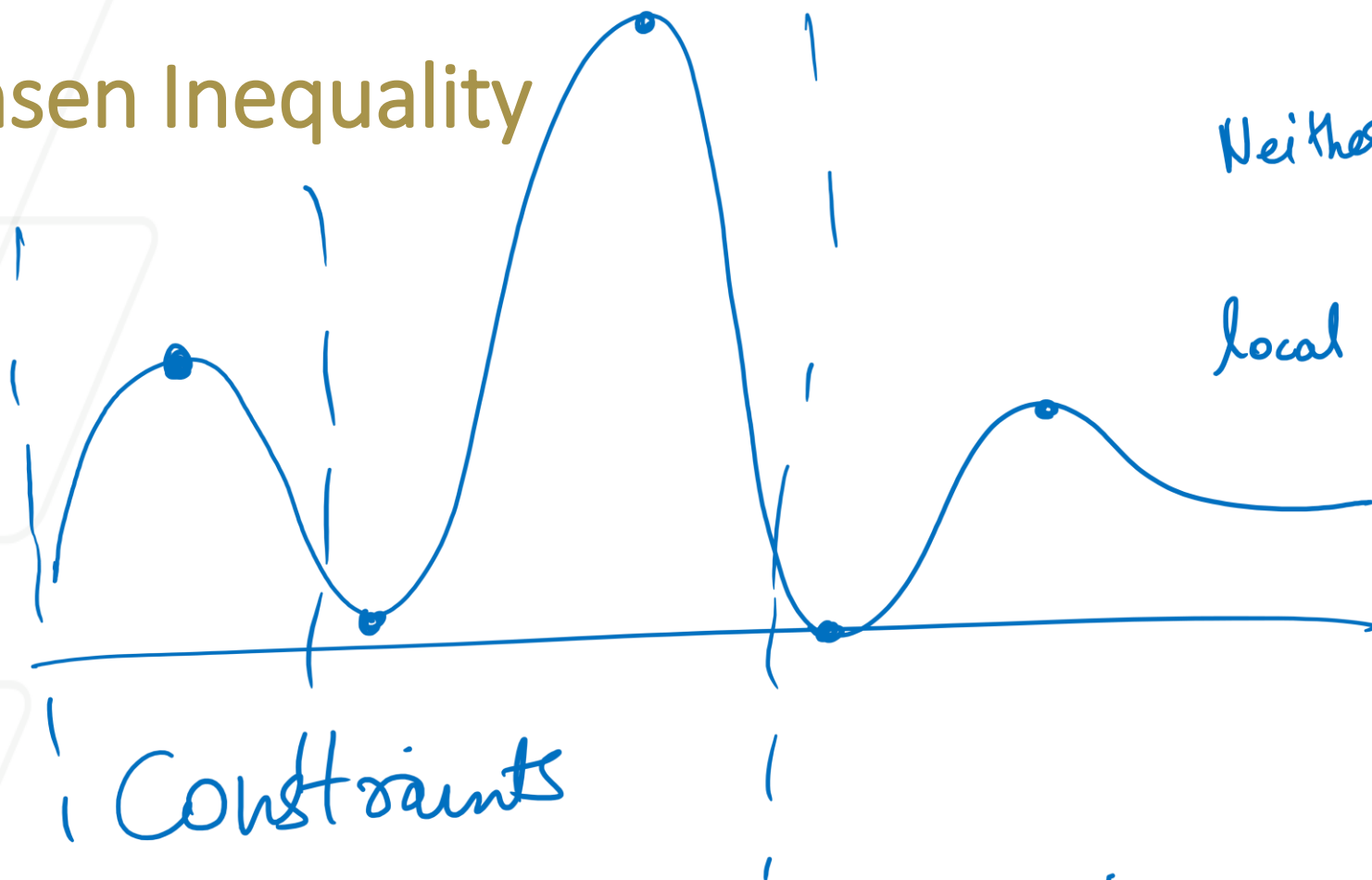
$$f(\mathbb{E}[x]) \quad \mathbb{E}[x] = \sum_x x \cdot p(x) = 1 \times \frac{1}{2} + 2 \times \frac{1}{2} = 1.5$$

Strictly convex functions have global minimum

Jensen Inequality

Neither concave & convex

local maxima & minima



$$\log(x) =$$
$$f'(x) = 1/x$$
$$f''(x) = -1/x^2$$



Concave.

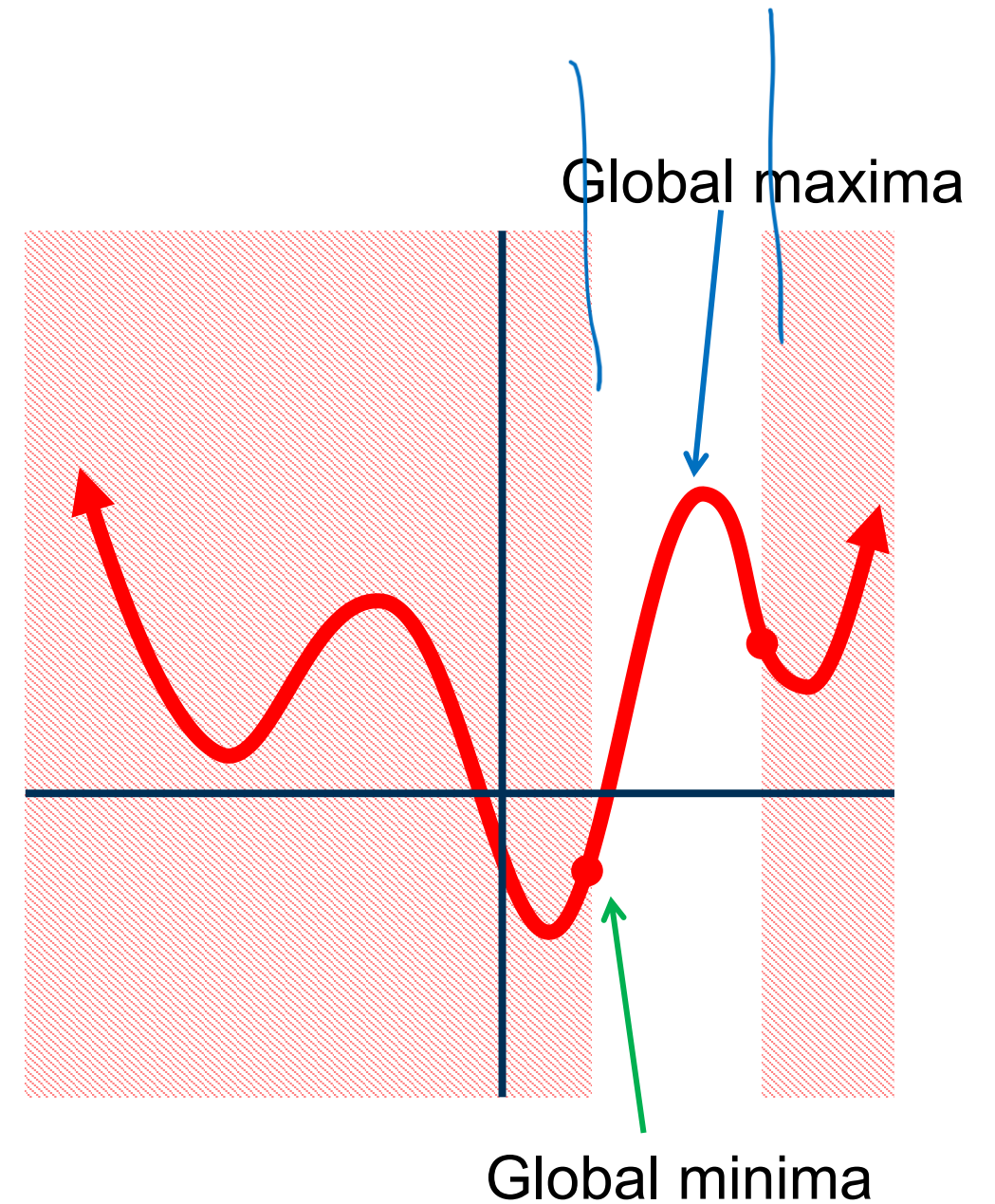
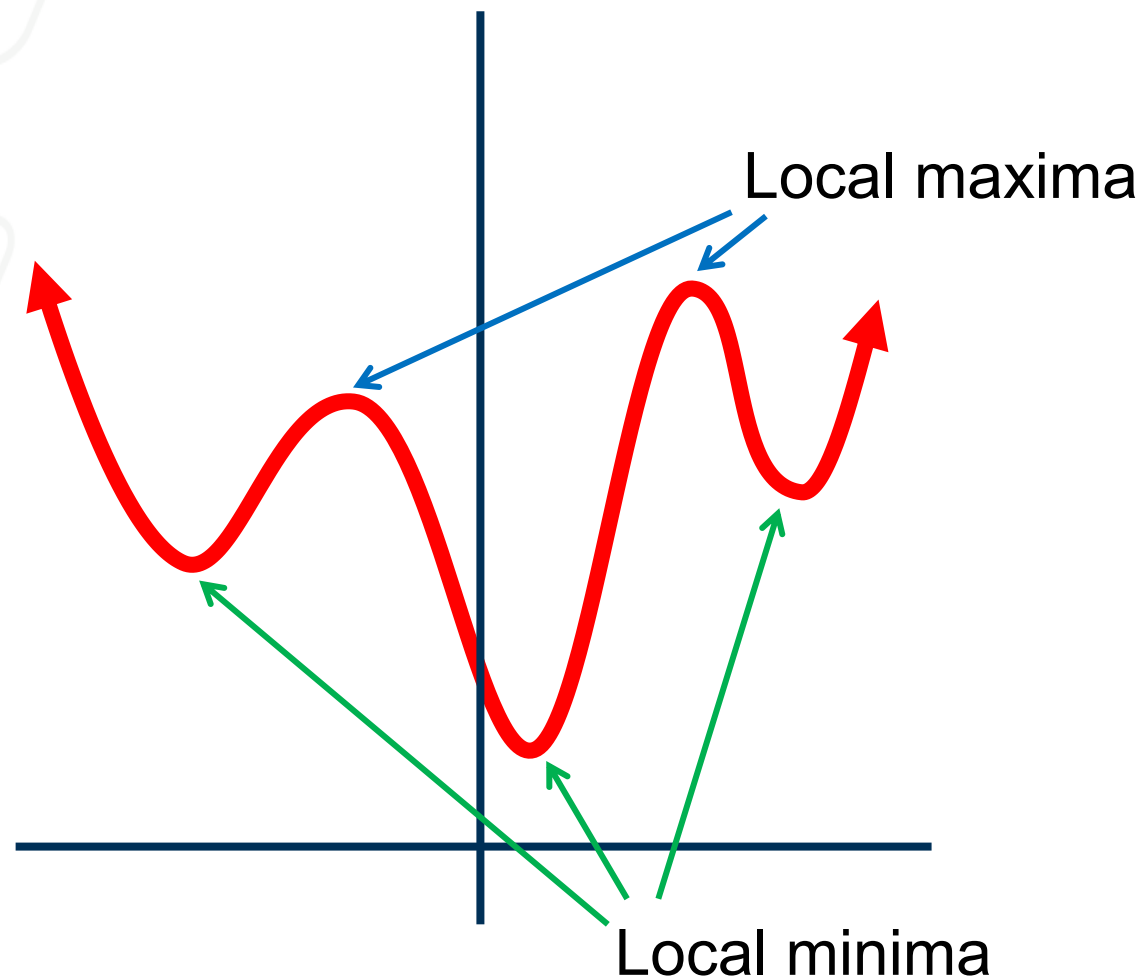
$$E[f(x)] \leq f[E(x)]$$

$$E[\log x] \leq \log(E(x))$$

KL Divergence is always non negative

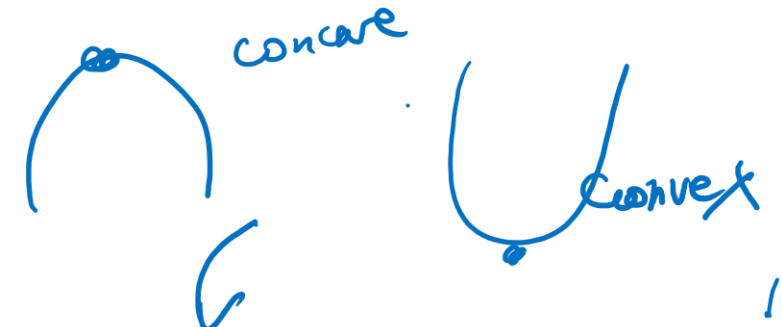
$$\begin{aligned}
 -KL[P][Q] &= \sum_x p(x) \log \frac{q(x)}{p(x)} && \xrightarrow{q(x)} \\
 &= \sum_x p(x) g(x) && \\
 &= E[g(x)] && \\
 &= \log(E[q(x)]) && \\
 &\leq \log\left(\sum_x p(x) g(x)\right) && \\
 &\leq \log\left(\sum_x p(x) \cdot \frac{q(x)}{p(x)}\right) && \text{pdf} \\
 &\leq \log \sum_x q(x) && \\
 &\leq \log 1 && \\
 &= 0 && \\
 -KL[P][Q] &\leq 0 && \\
 KL[P][Q] &\geq 0 &&
 \end{aligned}$$

Unconstrained and constrained optimization



$$f(x,y) = 6x^2 + 5y^2 \quad \text{Hessian} \begin{bmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial x \partial y} & \frac{\partial^2 f(x,y)}{\partial y^2} \end{bmatrix} \quad \det(\text{Hessian}) < 0 \rightarrow \text{Concave} \\ > 0 \rightarrow \text{Convex}$$

Basic unconstrained optimization problem



- Objective $f(\mathbf{x})$ the quantity we are trying to optimize (maximize or minimize)
- The variables $x_1, x_2, \dots, x_m$ which can be represented in vector form as $\mathbf{x}$
(Note: the subscript m indicates a feature NOT a datapoint in our dataset)
- Finding the stationary points where the function is no longer increasing or decreasing

Equality constrained optimization

$$\min f(x, y) = \frac{6x^2 + 5y^2}{\text{Convex}}$$

→ s.t. $x + y = 1$

- An equality constrained optimization problem is expressed as:

$$\begin{array}{ll} \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) & \text{objective} \\ \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) = 0} & \text{s.t.} \end{array}$$

$\underline{g(x, y) = x + y - 1 = 0}$

Inequality constrained optimization

- An inequality constrained optimization problem is expressed as:

$$\begin{array}{ll} \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \\ \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) \leq 0} \end{array}$$

K.K.T. conditions

Linear, Quadratic, and Nonlinear Programming

Linear Programming (LP)

Objective: Linear function of x .

Constraints: Linear.

Example: $\min 3x_1 + 5x_2 \text{ s.t. } x_1 + 2x_2 \leq 6$

Quadratic Programming (QP)

Objective: Quadratic

Constraints: Linear.

Example: $\min x^2 + y^2 + 3x + 5y \text{ s.t. } \underline{x + y \leq 4}$

Nonlinear Programming (NLP)

Objective: Nonlinear.

Constraints: Can be nonlinear.

Example: $\min x^2 + y^2 + 3x + 5y \text{ s.t. } x^2 + y \leq 4$



Recap

amt. of information we lose by predicting/using pdf $q(x)$ instead of original pdf $p(x)$

$$\rightarrow \sum p(x) \log q(x) = H[P] + KL[P][Q]$$

- Cross Entropy

//

minimizing CE is same as minimizing -ve log likelihood

- Loss Functions

// in a classification problem, CE as my loss/objective function, we are performing MLE.

CE loss function is very good because it penalizes divergence more & so convergence is quicker

- KL Divergence

↖

distance

measurement between 2 dist. $KL[P][Q] = -\sum_x p(x) \log \frac{q(x)}{p(x)}$

$$KL[P][Q] \geq 0$$

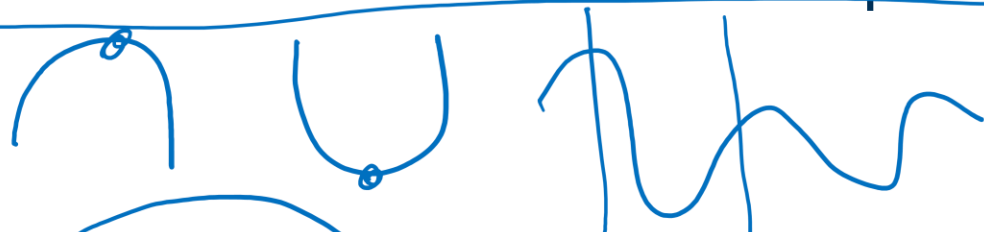
- Jensen Inequality



$$E[f(x)] \leq f(E[x])$$

$$E[f(x)] \geq f(E[x])$$

- Unconstrained and Constrained Optimization



- Linear, Quadratic and Non-Linear Programming

$f(x)$ is actually a pdf $\rightarrow$ MLE

$f(x) \leftarrow$ objective function
s.t. $g(x) \leq 0$

Equality constrained optimization problem

There is a notion of a Lagrangian expression where we combine both the objective function and our constraints

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x})$$

$$s.t. \lambda \neq 0$$

Equality constrained optimization problem

$f(x) \leftarrow$ objective function $\rightarrow$ unconstrained optimization $\frac{\partial f(x)}{\partial x} = 0 \quad x^*$

$f(x,y) = x^2 + y^2$
min.

s.t. $x+y=24$

$\frac{\partial f}{\partial x} = 0 \quad \frac{\partial f}{\partial y} = 0 \Rightarrow x, y = 0, 0.$

- With multiple equality constraints the Lagrangian is expressed as follows:

Lagrangian function $\rightarrow L(\mathbf{x}, \lambda) = \underbrace{f(\mathbf{x})}_{\text{objective}} - \sum_{i=1}^k \underbrace{\lambda_i}_{\text{Lagrange multiplier}} \underbrace{g_i(\mathbf{x})}_{\text{constraint}}$ s.t. $g(x) = 0$

s.t. $\lambda_i \neq 0$

$f(x,y) = x^2 + y^2$

s.t. $x+5y-1=0$

$g(x,y) = x+5y-1=0$

$L(x,y,\lambda) = x^2 + y^2 - \lambda(x+5y-1)$

Equality constrained optimization problem

$$L(x, y, \lambda) = x^2 + y^2 - \lambda(x + 5y - 1) = f(x, y) - \lambda(g(x, y))$$

$$\frac{\partial L}{\partial x} \text{ \& \& } \frac{\partial L}{\partial y} \qquad \nabla L = \nabla f - \lambda \nabla g = 0 \Rightarrow \nabla f = \lambda \nabla g$$

- To solve this optimization problem, we only need to satisfy the stationarity condition, which states:

$$\nabla L(\mathbf{x}, \lambda) = 0 \qquad \frac{\partial L}{\partial x} = 0$$

- We also need to satisfy

$$\lambda_i \neq 0$$

- This appeals to the intuition that at the optimal point, the gradient of the equality constraint is proportional to the gradient of the objective function

Equality constrained optimization: *Example 1*

$$L(x, y, \lambda) = x^2 + y^2 - \lambda(x + 5y - 1)$$

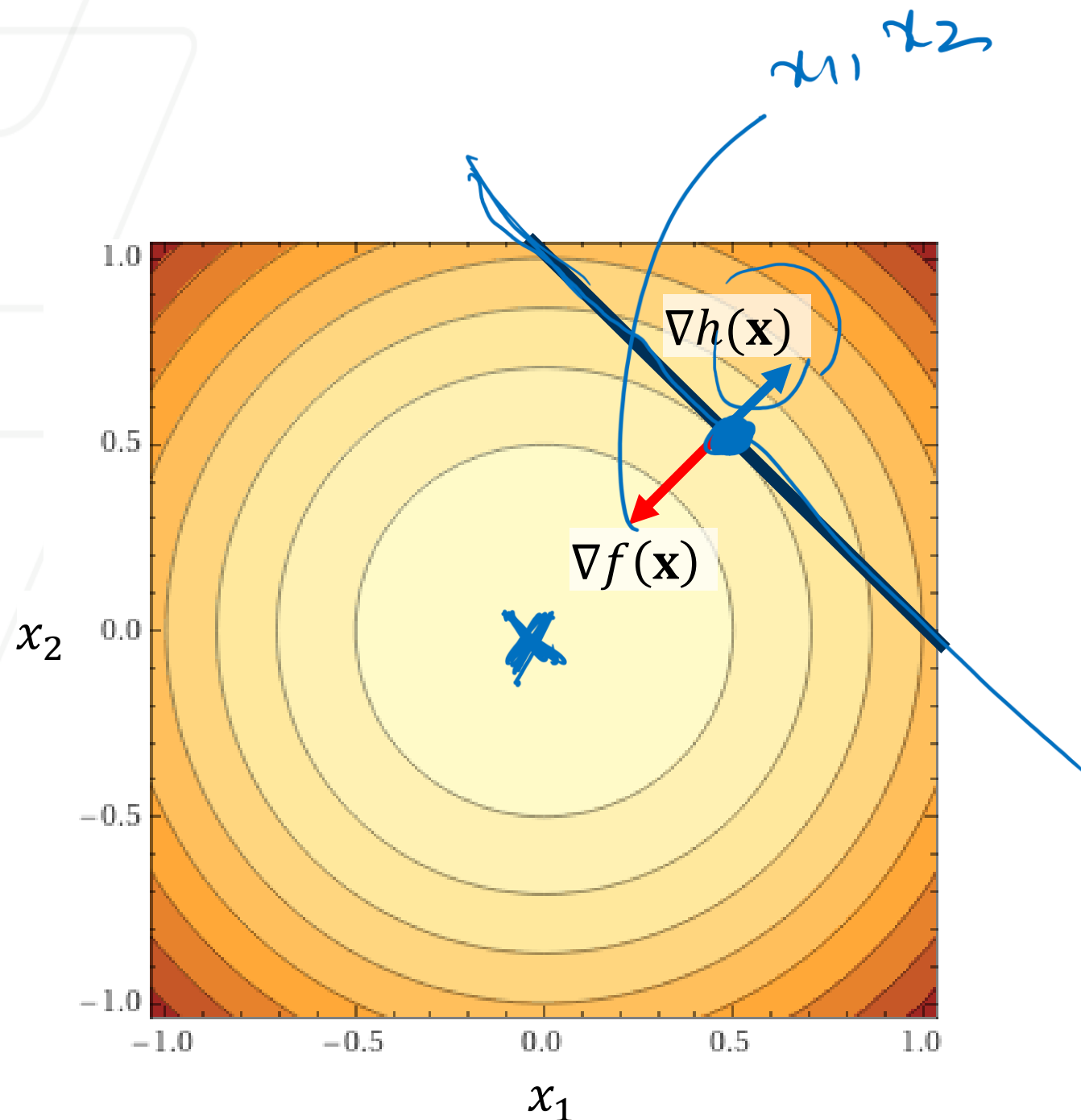
$$\frac{\partial L}{\partial x} = 0 \quad 2x - \lambda = 0 \Rightarrow x = \lambda/2$$

$$\frac{\partial L}{\partial y} = 0 \quad 2y - 5\lambda = 0 \Rightarrow y = 5\lambda/2$$

$$x + 5y - 1 = 0 \Rightarrow \frac{\lambda}{2} + \frac{25\lambda}{2} - 1 = 0 \Rightarrow \lambda = 1/13$$

$$x = \lambda/2 = 1/26, \quad y = 5\lambda/2 = 5/26$$

Equality constrained optimization: *Example 2*



$$\begin{array}{ll} \max_{\mathbf{x}} & \underbrace{1 - x_1^2 - x_2^2}_{\text{concave}} \\ \text{s.t.} & x_1 + x_2 = 1 \end{array}$$

$\downarrow$

$$\min_{\mathbf{x}} \quad -1 + x_1^2 + x_2^2$$

Equality constrained optimization: *Example*

$$\begin{array}{ll} \max_{\mathbf{x}} & 1 - x_1^2 - x_2^2 \\ \text{s.t.} & x_1 + x_2 = 1 \end{array}$$

$$\begin{aligned} & -f(x) - \lambda g(x) \\ & -1 + x_1^2 - x_2^2 - \lambda(x_1 + x_2 - 1) \end{aligned}$$

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = 1 - x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to $\mathbf{x}$

$$\frac{\partial L}{\partial x_1} = -2x_1 + \lambda = 0 \rightarrow \lambda = 2x_1$$

$$\frac{\partial L}{\partial x_2} = -2x_2 + \lambda = 0 \rightarrow \lambda = 2x_2$$

Equality constrained optimization: *Example*

3. Solve for $\mathbf{x}^*$ and λ

Using our previous two derivatives and the equality constraint, we obtain:

$$x_1 + x_2 = 1 \rightarrow x_1 + x_1 = 1 \rightarrow x_1^* = x_2^* = \frac{1}{2}$$

From $\frac{\partial L}{\partial x_1} = 0$ we find the value of λ

$$\lambda = 2x_1 \rightarrow \underline{\lambda = 1}$$

$$L(x, \lambda) = f(x) - \lambda g(x)$$

Inequality constrained optimization problem

- Constraints that limit how small or big variables can be. These are inequality constraints noted as $g_j(\mathbf{x})$
- An optimization problem is usually expressed as:

$$\begin{array}{l} \min_{\mathbf{x}} f(\mathbf{x}) \\ \text{subject to } g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m \end{array}$$

$\underbrace{g_j(\mathbf{x}) = 0}_{\text{equality constrained}} \rightarrow \text{Active Solution}$
 $\underbrace{g_j(\mathbf{x}) < 0}_{\text{Inactive Sol.}}$

Inequality constrained optimization problem

- The optimization problem can be rewritten as

$$\begin{aligned} & \min_{\mathbf{x}} \quad f(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x}) \\ & \text{subject to} \quad \lambda \geq 0 \end{aligned}$$

why?

- We need to satisfy the Karush-Kuhn-Tucker (KKT) conditions on $\mathbf{x}$, λ which is **necessary and sufficient** for optimality

Karush-Kuhn-Tucker Conditions

$$f(x,y) = x^2 + y^2$$

$$\text{s.t. } x + 5y - 1 \geq 0$$

$$g(x,y) = 1 - x - 5y \leq 0$$

- Stationarity condition:

$$\min L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x})$$

- Primal feasibility

$$g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m$$

$$g(x) \leq 0$$

- Dual Feasibility

$$\lambda_j \geq 0$$

$$\lambda \geq 0$$

- Complementary slackness

$$\lambda_j g_j(\mathbf{x}) = 0, j = 1, \dots, m$$

Case 1:

$$\lambda = 0, g(x) < 0$$

Case 2

$$\lambda > 0, g(x) = 0$$

If $g(x) < 0 \rightarrow$ inactive solution since $\lambda = 0$

If $g(x) = 0 \rightarrow$ active solution since λ is at play and > 0 , which means constraint is active

active sol.

Karush-Kuhn-Tucker Interpretations

- The complementary slackness condition applies only to inequality constraints
- It states that for a given inequality constraint...
$$\dots \text{either } g_j(x) = 0 \text{ or } \lambda_j = 0$$
- Whenever the dual variable $\lambda_j > 0$ and therefore $g_j(x) = 0$, we say that the **constraint is active** or that the **constraint is tight at $\mathbf{x}$**
- The reverse $\lambda_j = 0$ and therefore $g_j(x) < 0$ means that the constraint is inactive, meaning that it is not effectively impacting the solution

Rule: For each inequality constraint, **either** it is active and has a positive multiplier, or it is inactive and its multiplier is zero.

Primal vs. Dual Form

$$f(x, y) = \frac{x^2}{2} + \frac{y^2}{2} \quad \text{s.t.} \quad x+y=24$$

$$L(x, y, \lambda) = f(x, y) - \lambda(g(x, y))$$
$$= \frac{x^2}{2} + \frac{y^2}{2} - \lambda(x+y-24)$$

L is a function of x, λ

↓
primal form
original optimization
problem

$$\begin{cases} \frac{\partial L}{\partial x} = 0 & \frac{2x}{2} - \lambda = 0 \Rightarrow x = \lambda \\ \frac{\partial L}{\partial y} = 0 & \frac{2y}{2} - \lambda = 0 \Rightarrow y = \lambda \end{cases}$$

$$L(\lambda) = \frac{\lambda^2}{2} + \frac{\lambda^2}{2} - \lambda(\lambda + \lambda - 24) = \lambda^2 - 2\lambda^2 + 24\lambda = -\lambda^2 + 24\lambda$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow -2\lambda + 24 = 0 \Rightarrow \lambda = 12$$
$$x = 12$$
$$y = 12$$

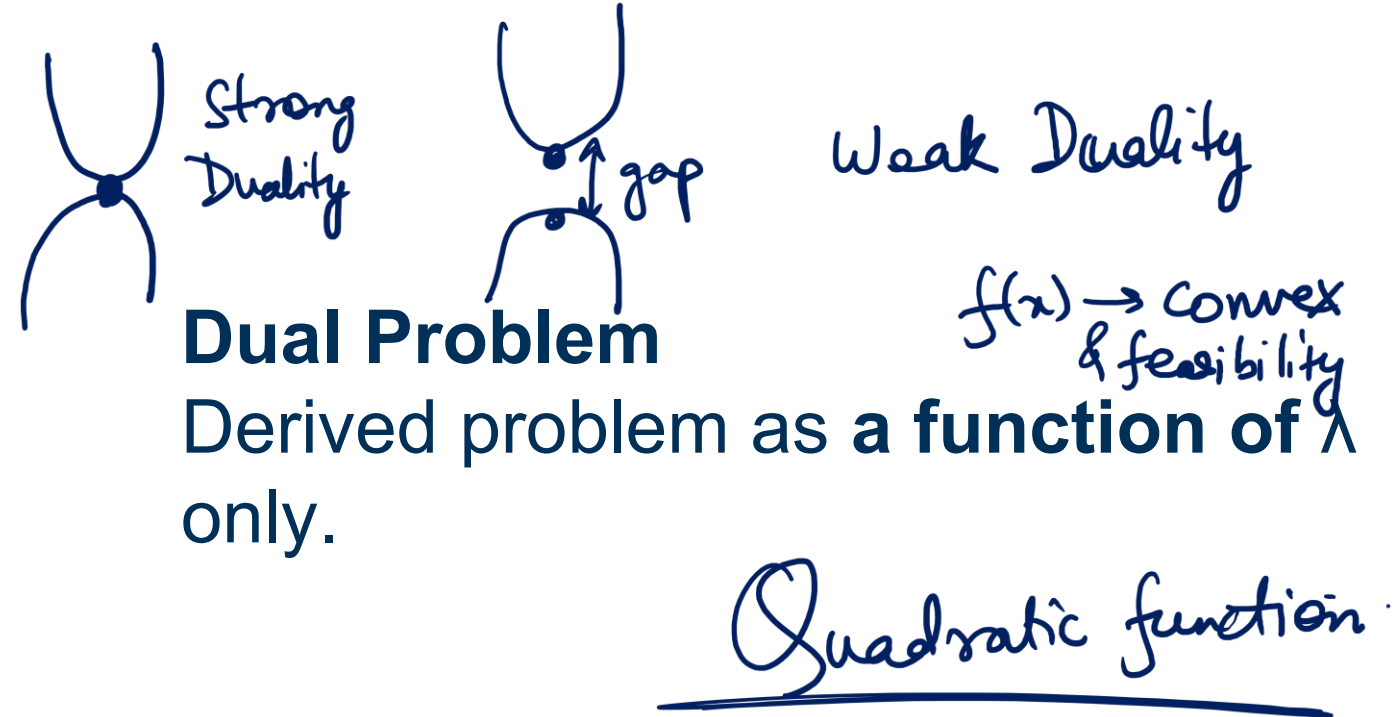
Dual.

Sol. of dual & primal are equal $\rightarrow$ Strong Duality
Sol. of dual $\leftarrow$ lower bound of primal $\rightarrow$ Weak Duality.

Primal vs Dual Form

Primal Problem

Original optimization problem



Analogy:

"Instead of directly solving for x and y together, we solve for what x and y *should look like* given a certain λ (like a candidate penalty price). Then we search over λ to find the best price that satisfies the constraint. This flips a 2D search into a 1D search."

Motivation:

In big problems (many x 's), solving for x in terms of λ can be done analytically or numerically, but then the dual problem might be much lower-dimensional $\rightarrow$ computationally cheaper.

Primal vs Dual Form

Weak Duality

- Dual optimum $\leq$ primal optimum (for minimization problems).
- Always true.

Strong Duality

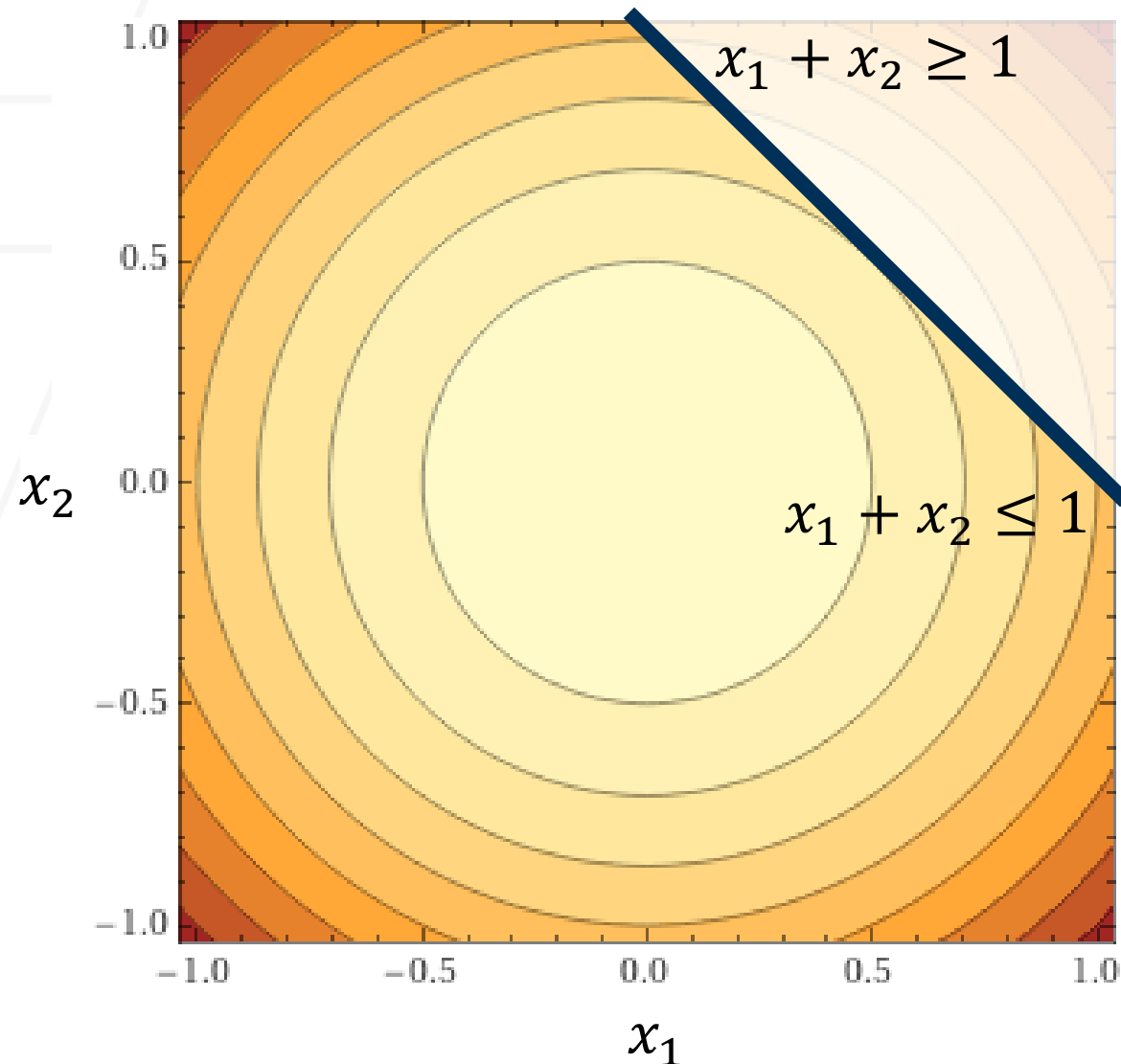
- Dual optimum = primal optimum.
- Holds for **convex problems** with feasible solution.

This is why **Quadratic Programming** is popular: QP is convex and has efficient solvers that work on the dual.

Why care?

Many ML models (e.g., SVMs) are solved via the dual.

Inequality constrained optimization: *Example 1*



$$\begin{array}{ll} \overset{\text{min}}{\max}_{\mathbf{x}} & 1 - x_1^2 - x_2^2 \\ \text{s.t.} & x_1 + x_2 \leq 1 \end{array}$$

Concave

$$f(x_1, x_2) = 1 - x_1^2 - x_2^2$$

$$g(x_1, x_2) = x_1 + x_2 - 1 \leq 0 \quad \checkmark$$

Stationarity
condition
min

$$\begin{aligned} \mathcal{L}(x_1, x_2, \lambda) &= -f(x) + \lambda g(x) \\ &= \underbrace{(x_2^2 + x_1^2 - 1)}_{\text{convex}} + \lambda \underbrace{(x_1 + x_2 - 1)}_{\text{-ve or 0}} \end{aligned}$$

Inequality constrained optimization: *Example 1*

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = -1 + x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to $\mathbf{x}$ (stationarity condition)

$$\frac{\partial L}{\partial x_1} = 2x_1 + \lambda = 0 \rightarrow \lambda = -2x_1$$

$$\frac{\partial L}{\partial x_2} = 2x_2 + \lambda = 0 \rightarrow \lambda = -2x_2$$

$$x_1 = x_2$$

$$\lambda \geq 0$$

$$g(x) \leq 0$$

Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

Case 1: $\lambda > 0$ and
 $g(x) = 0$

$$\underline{x_1 + x_2 - 1 = 0} \rightarrow 2x_1 = 1$$

$$\underline{x_1 = x_2 = \frac{1}{2}}$$

$$\lambda \geq 0$$

4a. Check if the dual feasibility condition is satisfied

$$\lambda = -2x_1 = -1$$

$$\lambda = -1$$

This solution **does NOT** satisfy the dual feasibility condition and it is thus **unfeasible**

Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

Case 2: $\lambda = 0$ and $g(x)$
 < 0

$$\underline{\lambda = -2x_1}$$

$$\underline{x_1 = x_2 = 0}$$

4b. Check if the primal feasibility condition is satisfied:

$$\underline{g(x_1, x_2) = 0 + 0 - 1 = -1 < 0}$$

The solution is feasible and therefore the optimal!

Inequality constrained optimization: *Example 1*

Summary of the KKT conditions for the problem:

$$\left. \begin{aligned} \lambda &= -2x_1 \\ \lambda &= -2x_2 \end{aligned} \right\} \text{Stationarity}$$

$$x_1 + x_2 - 1 \leq 0 \quad \text{Primal feasibility}$$

$$\lambda \geq 0 \quad \text{Dual feasibility}$$

$$\lambda(x_1 + x_2 - 1) = 0 \quad \text{Complementary slackness}$$

not closed form solution.

$$f(x) = e^x + 2x$$

Gradient descent

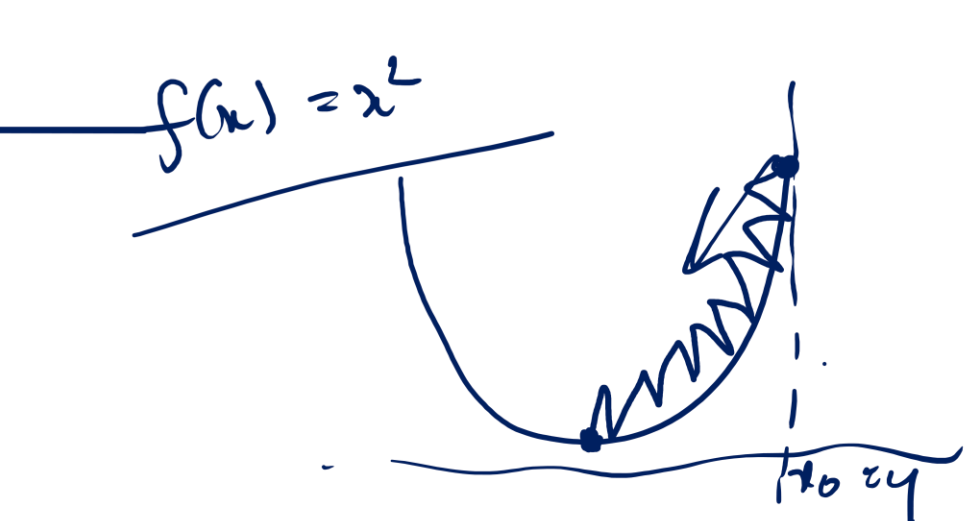
concave



gradient ascent:

$$x_{n+1} = x_n + \alpha \nabla f(x_n)$$

- In machine learning, some of the optimization problems suffer from the following challenges:
 - A closed form solution may be computationally intractable
 - Functions are complex: no easy derivative = 0 solution.
 - High dimensions: solving $\nabla f(x)=0$ analytically is impossible.
 - All the data may not be available at the time of training (e.g. streaming data)
- Gradient descent is an iterative minimization technique for differentiable functions on a domain
- **Intuition:** the function $f(\mathbf{x})$ decreases the fastest by going from x_n to x_{n+1} in the opposite direction of the gradient of $f(\mathbf{x})$

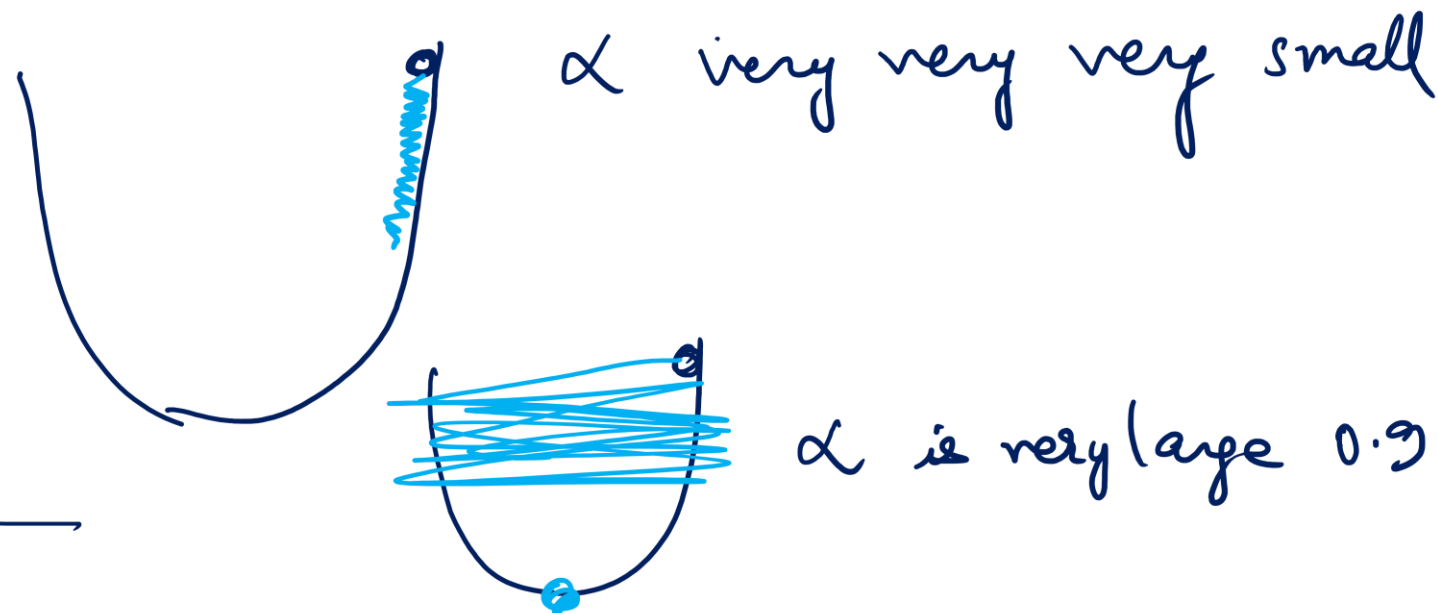


$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha_n \nabla f(\mathbf{x}_n)$$

learning rate = 0.000625

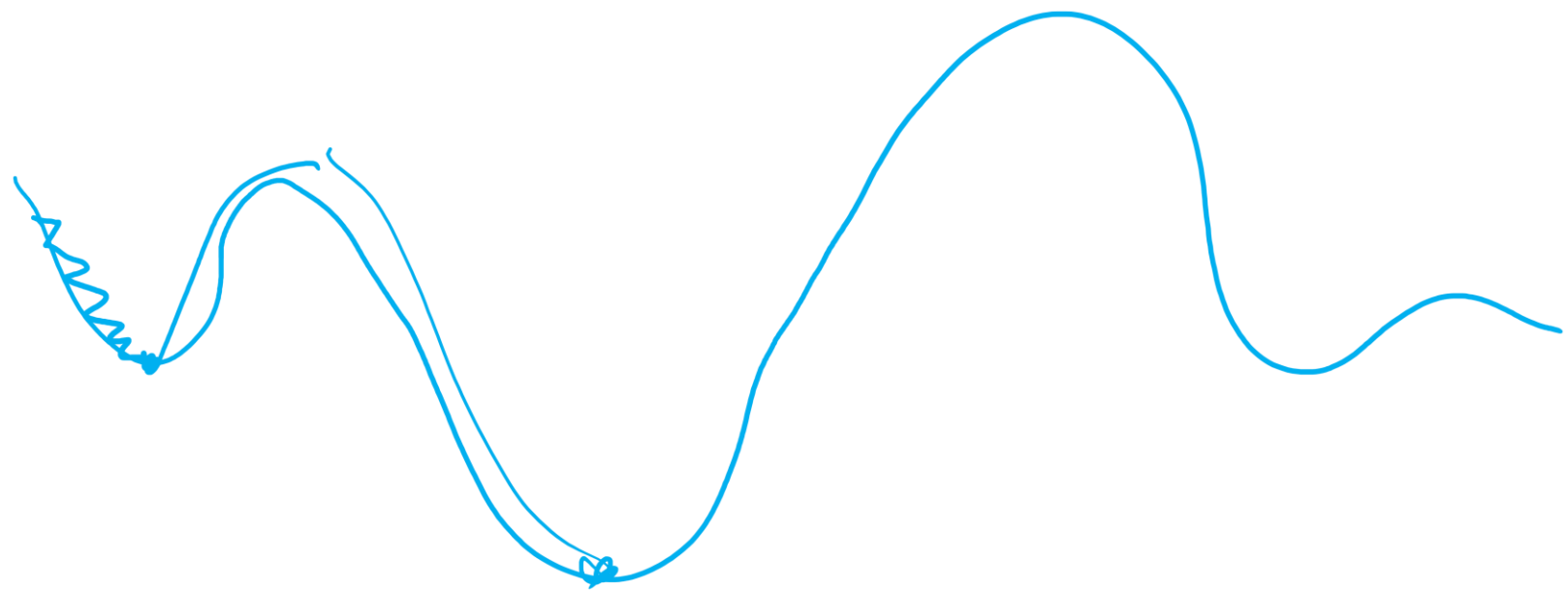
$$x_1 = 4 - 0.000625 \cdot 2x$$

Learning rate



- The choice of the learning rate α_n plays an important role in the success of the minimization:
 - If α_n is **too small** then the number of iterations necessary to reach the minimum will be too large
 - If α_n is **too large** then we may overshoot the minimum
- It can be set to a constant or varied iteratively to improve the performance of the algorithm

Momentum



- The gradient descent technique is very susceptible to local minima, therefore, it can be modified by adding a momentum term

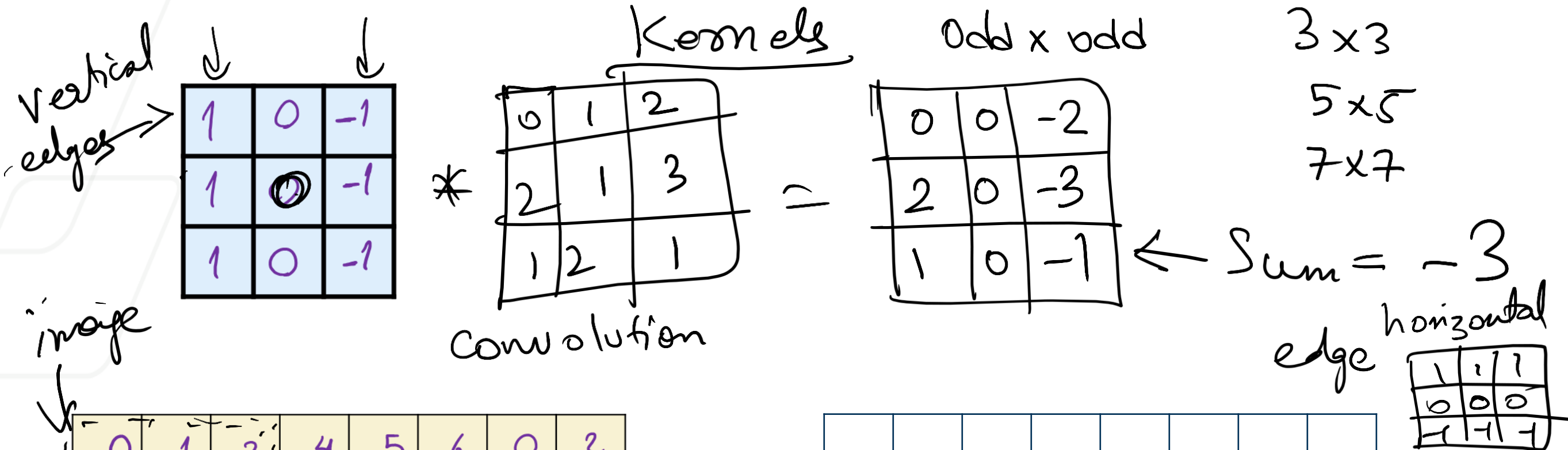
$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha_n g(\mathbf{x}_n)$$

Where:

$$g(\mathbf{x}_n) = \mu g(\mathbf{x}_{n-1}) + \nabla f(\mathbf{x}_n)$$

μ = momentum constant

- The momentum constant also needs to be fine tuned along with the learning rate to each problem, typically picked as a value between 0 and 1



0	1	2	4	5	6	0	2
2	1	3	8	9	255	72	83
1	2	1	4	79	65	53	33
43	97	15	67	104	77	163	43
36	173	13	76	205	89	179	34
63	163	113	86	209	91	185	98
84	153	123	96	134	101	196	121
96	143	133	79	135	103	216	211

	-3						

↑ grayscale → 0 - 255